

Dragonfly Health

ADT Interface Specifications

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ADT

1. INTRODUCTION

Dragonfly Health stores patient information by billing group, as defined by the financial agreement between the hospice and Dragonfly Health. The functionality of the inbound ADT interface depends on Dragonfly Health’s ability to identify transactions sent to the pharmacy system. To support this, Dragonfly Health may use the patient ID generated by the EMR, along with additional identifiers, to accurately identify the patient. When sufficient identifying information is provided, interface transactions will be processed accordingly. This specification document and the accompanying sample messages are universal and apply to HL7 2.x, XML, JSON, and FHIR JSON R4 formats.

2. DOCUMENT CONCEPTS

2.1 Definitions

- Inbound ADT Interface - ADT messages transmitted from the client’s EMR system to Dragonfly Health
- Transmission method – Push method – EMR sends ADT messages to Dragonfly Health
- Interface Engine – Dragonfly Health’s proprietary interface engine (the SIG – Single Interface Gateway) that manages transactions and validation between Dragonfly Health and client’s hospice management system.
- End users - client users of Dragonfly Health’s systems and their EMRs.
- Dragonfly Health Systems include client-facing systems such as:
 - E3Pro App
 - DMETTrack
 - Dragonfly Health Call Center
- EMR system hosting
 - Client hosted – each client hosts their own patient database.
 - Vendor hosted - the EMR/vendor hosts patient databases for all clients.

3. PURPOSE OF THIS DOCUMENT

To provide Dragonfly Health hospice partners and vendor agencies with a detailed description of the messages used in the Dragonfly Health interface. This includes messages inbound to Dragonfly Health for Demographic (Referral, Admit, Update, Discharge and Transfer) transactions. This document will describe the format of those messages and field requirements.

3.1 Intention of the Dragonfly Health Interface Engine – Demographic (Referrals, Admissions, Discharges, Transfers)

In general, an interface transaction sent to Dragonfly Health will process correctly only if all the following conditions are met:

- All the required fields for the interface are present and formatted correctly.
- The data follows all current Dragonfly Health business rules relating to census management and billing.

If patient data is entered into Dragonfly Health through resources outside of the interface (i.e. E3Pro, DMETTrack, or the call center) prior to the initial demographic interface transaction, the interface transaction might create a duplicate patient profile or fail. Patient information may be modified by the hospice via a phone call to Dragonfly Health or by direct web access via Dragonfly Health Online Tools. Contents of the referral, admission, update, discharge and transfer messages are described in more detail in section 4.

Please note that any changes made to a patient within Dragonfly Health may be overwritten by the interface, if the corresponding change is not first made within the vendor system. Clients are encouraged to make changes via the interface only.

4. DRAGONFLY HEALTH BUSINESS RULES AND EXPECTATIONS

All the message types have the same field requirements. The only exceptions are discharge messages and discharge dates. Dragonfly Health is not expecting discharge dates in any other message types. If a transaction is sent to Dragonfly Health that does not meet the requirements, the transaction will not be processed.

4.1 Org/Agency identifiers and mapping elements

- Dragonfly Health uses agency identifiers to process and route messages appropriately within Dragonfly Health's systems.
- Dragonfly Health will share the organization ID for each agency/branch at the start of the implementation process. It is expected that this identifier will be sent in all ADT transactions. Alternatively, Dragonfly Health can also accept EMR vendor's organization identifiers.
- It is expected that the EMR vendor will share the branch/agency, location, and team identifiers (if applicable) at the start of the implementation process. It is expected that these identifiers will be sent in all ADT transactions.
- Dragonfly Health and the EMR vendor and/or hospice client must come to an agreement on the exact definitions of the hierarchical hospice identifiers.
- The hierarchy of location identifiers is as follows:

**NOTE: DMETrack does not utilize group, team, or level of care*

- Parent/Master org – corporate office of an organization (hospice) with multiple locations
- Org (organization) / Agency / Branch – a location under a parent / master org
 - An organization with a single location will only utilize the org level concept, it will not utilize the parent/master org concept.
- Group / Team / Location / Level of Care / Place of Service – a sub location under an org.
 - It is expected that the locations (such as: Home, ALF, LTC) are standard across agencies. If the locations are not standard, they need to be defined early in the project.
 - It is expected that the same team will NOT be under two different groups or locations.
 - The team concept needs to be defined early in the project (if applicable)

4.2 Patient Identifiers

- Medical Record Number (MRN)
 - MRNs are expected to be the patient's primary identifier.
 - MRNs are required for all messages including referrals.
 - It is expected that MRNs are AUTO generated in the EMR and cannot be changed. The EMR vendor needs to provide the schema for the MRN creation.
 - If MRNs are NOT AUTO generated, a detailed rationale and workflow is required.
 - It is expected that a hospice cannot have duplicate MRNs for different patients within the same database for all the branches.
- Social Security Number (SSN)
 - The Dragonfly Health pharmacy system does accept SSN as a patient identifier, however MRN is preferred. The DME system does NOT accept SSN as a patient identifier.
- If there are any other patient identifiers (such as episode IDs, patient admit IDs, internal patient IDs), that need to be included in the ADT interface, such concepts need to be discussed early in the project. If there are any client specific workflows regarding patient IDs, such concepts need to be discussed early in the project.

4.3 Referral

- Hospices may send referral transactions, pre-admits.
- If the Referral patient is not activated (admitted) within 30 days in the pharmacy system, the patient information is automatically archived in.
- If an Active patient does not have any equipment ordered within 30 days in the DME system, the patient is set to Inactive.
- If a referral transaction is sent after a patient has been activated (admitted), the transaction is not processed as a referral.
- It is assumed that a referral action by a clinician within the EMR/vendor system will trigger an outbound referral message to Dragonfly Health.
- Dragonfly Health will process all referrals as referrals in the pharmacy system and as admits in the DME system. Any other expectations of referral message behavior should be discussed in advance.
- The following elements are NOT required in a referral message:
 - The patient's primary provider's NPI
 - Diagnosis codes and allergies
- Referral date (Period Start Date) is a required element in a referral message.

4.4 Admission / Registration

- Once the admission message is received through the interface the patient is admitted into Dragonfly Health's pharmacy system and/or DME system depending on the client contract and configuration.

- If the patient has been admitted to Dragonfly Health via our online tools or our call center, the subsequent interface admit transaction might create a duplicate patient profile or fail. Clients are encouraged to admit patients via the interface only.
- Pharmacy patient admission transactions are allowed with an admit date of 15 days in the past and 3 days in the future of the current date. The DME system does NOT have any admit date rules.
- Allergies are expected in admission messages and/or subsequent update messages for pharmacy.
 - Allergy is a repeatable node in a patient ADT message.
 - If a patient does NOT have any allergies, Dragonfly Health is expecting an “NKA” (No Known Allergies) to be sent in the message. It is assumed that NKA is a selectable allergy choice and NOT a free text.
 - Admission messages without allergy information will NOT fail. Absence of allergy information might slow down or prevent medications from being dispensed.
 - Dragonfly Health expects clinicians to select allergies from a drop-down menu. In case of free text allergies or combination of drop-down and free text, the vendor and the client understand that some allergies might not be successfully loaded into Dragonfly Health’s pharmacy system.
 - Dragonfly Health can only add allergies to the patient’s profile. Allergies cannot be removed at this time. NKA (No Known Allergies) is the exception to the rule. If the admit messages contain “NKA” as an allergy, a subsequent message with an actual allergy will be processed and NKA will be replaced in Dragonfly Health’s pharmacy systems.
 - The allergy information is as follows:
 - Allergy Name / Description - REQUIRED
 - Onset Date
 - Codeable concept (if applicable) such as: SNOMED codes or NDC if med allergy
 - Allergy Status: Active or Inactive (FHIR only)
 - Allergy type (if applicable)
 - Medication
 - Food allergies
 - Environment
 - Biologic
- Diagnoses / ICD10s are expected in admission and/or subsequent update messages for both pharmacy and DME.
 - Diagnosis is a repeatable node in a patient ADT message.
 - If ICD10s are NOT sent, this might slow down or prevent medications from being dispensed.
 - Dragonfly Health is expecting only active diagnosis codes to be sent in each message. If a diagnosis code becomes inactive, it should not be included in the subsequent message.
 - The diagnosis information is as follows:
 - ICD10 code - REQUIRED.
 - ICD10 name / description
 - Primary diagnosis code indicator – REQUIRED.
 - Onset Date
 - Resolution Date
 - IDC10 status: Active or Inactive (FHIR only)
- Readmission
 - If a patient is readmitted after being admitted and discharged, Dragonfly Health expects the same MRN number to be used.
 - If a patient is readmitted, any patient demographic updates in the re-admit message will be reflected in Dragonfly Health’s system.
- If a concept of admission to an unspecified or pending group or team or location exists, such concepts need to be discussed and defined during the planning stage of the integration project.
- Admission Date (Period Start Date) is expected in an admission message. If the admission date is different than the Referral date, Dragonfly Health is expecting the true admit date in the admit message.
- For patients in an LTC and ALF location the internal vendor facility ID, facility name, address and phone number are expected in an admit message and subsequent update and transfer messages where applicable. Handling of LTC/ALF patients depends on the nuanced definition of facility and billing group and org configuration.

4.5 Discharge / Death

- The patient must be active (admitted) before a deceased discharge, or a live discharge transaction can be processed. A deceased discharge or live discharge transaction will fail if a patient is not admitted.
- The patient death indicator is the differentiator between a live discharge and deceased discharge. A blank death indicator will be processed as live discharge.

- Decease discharge and live discharge transactions must have a death/coverage stop date no more than 15 days in the past or 3 days in the future.
- A decease discharge message received after a live discharge message for the same patient will NOT be processed in the pharmacy system. Such patients have to be manually deceased by calling the DFH call center. A decease discharge will be processed after a live discharge in the DME system.

4.6 Updates/Changes

- Updates may be processed on patients who are active (admitted) or in referral status.
- The discharge date is the only updatable element for a discharged / deceased patient in the pharmacy system if the org is configured for this setting. Alternatively, the DME system accepts all updates post discharge / decease.
- The following elements are updateable fields for all patients:
 - First, Last, and/or Middle Name
 - Gender
 - Date of Birth
 - Allergies
 - Diagnoses
 - Address (only updateable in the pharmacy system for home patients)
 - Phone (only updateable in the pharmacy system for home patients)
- Within the pharmacy system, any address or phone number updates submitted via update transaction for patients residing in a facility (such as LTC or ALF) will NOT be processed. The patient's primary address will always match the facility address.

4.6 Transfers

- Patient needs to be in an active (admitted) status for interface transfers to process.
- The concept of interagency transfers does NOT exist in the DME system. All A02's received via the Interface for interagency moves will be treated as updates.
- The transfer date field must be clearly defined and transmitted in the Location Effective Date. The Location Effective Date (EVN.2) is not the same as Coverage Start Date (PV1.44).
- Patient transfers can have a Location Effective Date of no more than 15 days in the past and 3 days in the future of the current date.
- Same day transfers are supported by Dragonfly Health's interface.
 - However, the Dragonfly Health pharmacy system will add 1 (one) day to the Location Effective Date of the second transfer or the first transfer is the admit is on the same day. The end date of the first period and the start date of the second period cannot be the same in Dragonfly Health's system.
 - A discharge transaction on the same day as the two transfers will still process with the discharge date from the discharge message.
- Transfers from Facility to Home will update the patient's address and the Dragonfly Health pharmacy billing Group.
- Dragonfly Health is expecting transfer or update messages to be sent for transfers. Sending discharge messages and readmit messages to serve as transfers is not the expected behavior.
- Dragonfly Health only allows transfers between Dragonfly Health orgs under the same parent/master/company orgs.
- Transfer messages sent prior to the patient being in admitted status will fail.
- Transfers to the same group or team the patient is will not transfer the patient; any updated patient demographics will update the patient record.

5. ASSUMPTIONS

5.1 Assumptions:

- One patient will be transmitted per message.
- ADT messages will only contain the current coverage period. Previous coverage periods should not be included in the message. If multiple coverage periods are included in the message DFH will value the most recent coverage period.
- If an element value belongs to a codable concept set, the name of the codable concept should be sent.
 - Allergies, Race and Ethnicity
- No changes will be made to Dragonfly Health's front end systems.
- This document is for ADTs only. Medications are covered in a separate document.

- Connectivity – it is assumed that the vendor can send messages for multiple unrelated clients through the same channel.
- It is assumed that ADT messages are sent in a live environment as soon as each ADT action is completed. If there are any holding mechanisms, batch sends or offline modes that trigger mass ADT messages; such processes need to be described in detail ahead of time during testing and production configuration.
- Patients’ insurance information can be sent; however, it will not be consumed since most of Dragonfly Health’s clients are billed directly by Dragonfly Health. Patient insurance is a repeatable node in an ADT message.

It is assumed that EMR user actions (ACE rules) that trigger ADT messages to Dragonfly Health are standard across the EMR for all clients. All the trigger actions (ACE Rules) need to be defined. Additional definitions are necessary if the triggers are not standard.

6. MESSAGES

6.1 Message Format

- **JSON is the preferred format by Dragonfly Health for ADT and medication interfaces.**
- Dragonfly Health also accepts HL7 2.x, XML, and FHIR JSON (<http://hl7.org/fhir/>) formatted messages.

6.2 Message Types

- Message types - JSON standard messages do NOT contain an indicator for message type. The contents of the message will be interpreted to determine the status and actions to be taken to the patient’s records in Dragonfly Health’s system. HL7 2.x, XML and FHIR JSON messages need to contain the message type.
 - Referral
 - The concept of referrals does not exist in the DME system so any referral received via the interface will be processed as an admit into DMETrack.
 - Referrals will be processed as referrals in the pharmacy system.
 - Admission – Dragonfly Health will use the admission message received through the interface to admit the patient into our pharmacy and/or DME system.
 - Update – Dragonfly Health can accept any patient demographic update.
 - Transfer – patient transfer between branches, groups or locations. FHIR JSON specs do not implicitly identify a transfer message. Dragonfly Health will use any subsequent messages with an Active status as a transfer message if the branch, group or location in the message are different than those current values in the Dragonfly Health database. The current branch, group or location values would have been derived from the previous Active message or updated by a user in Dragonfly Health systems. If the patient demographics in the message are different than the patient demographics in the Dragonfly system, the information from the message will be used to update the patient record.
 - Discharge - Discharge transactions may be identified within the vendor system as the following: Finished.
 - Live discharge: XML Message Type = 18,
 - Deceased discharge: XML Message Type = 19, HL7 2.x Message Type A03, Death Indicator = PID.30
 - Other patient status message – OnHold, Cancelled or Entered-In-Error will be handled as discharges in Dragonfly Health’s systems. Should the patient need to be readmitted a new admit message with an Active status needs to be sent.

Message Type / Patient Status Reference Table				
	HL7 2.x (MSH.9.2)	XML (RecType)	JSON	FHIR JSON (patient status)
Referral	A05	99	n/a	planned waitlist onhold
Admit	A01, A04, A28	10	n/a	active
Update/Transfer	A02, A08, A31, A34, A60	8,14,17,20	n/a	active
Discharge	A03, A11	18,19	n/a	finished cancelled entered-in-error



6.3 Message Description

If the data below is filled out or populated by the user within the EMR/vendor system, it is expected that all the appropriate fields are populated in the outbound message to Dragonfly Health. These fields are minimally required; however minimal data will impact functionality in Dragonfly Health’s client facing applications.

Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
patient	ADT	--	EpisodeOfCare	--	R	Encapsulating field
n/a	RecType	MSH.9.2	EpisodeOfCare.status	XML: integer HL7: string FHIR JSON: code	R*	ADT request message type. *NOT required for JSON message type. (see reference table above in section 6.2)
Organization/Coverage						
--	--	--	Organization.id	FHIR JSON: id	R*	FHIR JSON resourceType container identifier *Only required for FHIR
institution.name	Hospice/HospiceName	PV1.3	Organization.name	string	R	Name of the hospice/branch
institution.id	Hospice/HospOrgCode	MSH.4	Organization.identifier	JSON: integer XML: integer HL7: integer FHIR JSON: Identifier* *NOTE: map to the value element & set system to the URI for identifier (specifying DFH org id)	R	Hospice ID assigned by Dragonfly Health
location	Hospice/HospLocType	PV1.4	EpisodeOfCare.type	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept* *NOTE: map to the coding.code element & set coding.system to the URI to specify the care setting type	R	Hospice Location / Care Setting Type -> Home ALF LTC IPU
--	--	--	CareTeam.id	FHIR JSON: id	R*	FHIR JSON resourceType container identifier *Only required for FHIR
group	Hospice/HospGroupID	--	--	string	O	Hospice Group name or ID assigned by Dragonfly Health
team_name	Hospice/HospTeamName	PV1.10	CareTeam.name	string	O	Hospice Team Name



ADT Interface Specifications

Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
location_effective_date	Hospice/HospLocEffDate	EVN.2	CareTeam.period.start	dateTime	R	Location Effective Date – this is used as the transfer date NOTE: Dragonfly Health utilizes a date-based billing cycle rather than time-specific increments.
pending_date (for referrals) or admit_date (for admitted patients)	Coverage/CovStartDate	PV1.44	EpisodeOfCare.period.start	dateTime	R	Patient Coverage Start Date Dragonfly Health is expecting this date to remain static for the duration of the patient episode (Admit to Discharge). NOTE: Dragonfly Health utilizes a date-based billing cycle rather than time-specific increments.
discharge_date	Coverage/ CovStopDate	PV1.45	EpisodeOfCare.period.end	dateTime	R*	Patient Coverage Stop Date *Only required in a discharge message NOTE: Dragonfly Health utilizes a date-based billing cycle rather than time-specific increments.
deceased	RecType 18 = live discharge RecType 19 = deceased	PID.30	Patient.deceasedBoolean	JSON: boolean XML: (based on RecType) HL7: char(1) – accepts Y or N FHIR JSON: boolean	R*	Patient Deceased Indicator - TRUE = deceased - FALSE = live discharge *Only required in a discharge message
Patient						
--	--	--	Patient.id	FHIR JSON: id	R*	FHIR JSON resourceType container identifier *Only required for FHIR
patient_id	Patient/PatID	PID.3.1	Patient.identifier	JSON: string XML: string HL7: string FHIR JSON: Identifier* *NOTE: map to the value element & set system to the URI for identifier (specifying 'MRN')	R	Patient MRN
patient_ssn	Patient/PatSSN	PID.19	Patient.identifier	JSON: string XML: string HL7: string FHIR JSON: Identifier*	O	Patient SSN

Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
				<i>*NOTE: map to the value element & set system to the URI for identifier (specifying 'SSN')</i>		
last_name	Patient/PatLastName	PID.5.1	Patient.name.family	string	R	Patient last name
first_name	Patient/PatFirstName	PID.5.2	Patient.name.given[0]	string	R	Patient first name
middle_name	Patient/PatMiddleName	PID.5.3	Patient.name.given[1]	string	O	Patient middle name
street_address	Patient/PatAddress1	PID.11.1	Patient.address.line[0]	string	R	Street address of the patient
address2	Patient/PatAddress2	PID.11.2	Patient.address.line[1]	string	O	Second line of the street address of the patient
city	Patient/PatCity	PID.11.3	Patient.address.city	string	R	Patient city
state	Patient/PatState	PID.11.4	Patient.address.state	string	R	Patient state
zip	Patient/PatZip	PID.11.5	Patient.address.postalCode	string	R	Patient zip code
phone	Patient/PatHomePhone	PID.13	Patient.telecom	JSON: string XML: string HL7: string FHIR JSON: ContactPoint* <i>*NOTE: map to the value element, set system to 'phone' & use to 'home'</i>	R	Patient home phone number The phone number should NOT contain any parenthesis, space or dashes (e.g. 1112223333) NOTE: If no phone number is received Dragonfly Health will use the hospice phone number as the patient phone number
birth_date	Patient/PatBirthDate	PID.5	Patient.birthDate	date (YYYYMMDD)	R	Patient date of birth
sex	Patient/PatGender	PID.8	Patient.gender	string	R	Patient gender - M – male - F – female - U – unknown other (NOTE: FHIR JSON requires the full word, all lowercase)
--	--	--	Patient.active	FHIR JSON: boolean	R*	Patient active boolean. Indicates whether this patient's record is in active use. *Only required for FHIR JSON
Diagnoses (repeatable)						
--	--	--	Condition.id	FHIR JSON: id	R*	FHIR JSON resourceType container identifier *Only required for FHIR
diagnosis.code	Diagnosis/DiagICD10	DG1.3.1	Condition.code	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept*	P	Diagnosis ICD10 code While diagnoses are NOT required in any ADT messages, medications cannot be dispensed without an ICD10 present.



Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
				<i>*NOTE: map to the coding.code element & set coding.system to the URI to specify the code type</i>		
diagnosis.description	Diagnosis/DiagDescription	DG1.3.2	Condition.code	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept* <i>*NOTE: map to the coding.display element & set coding.system to the URI to specify the code type</i>	O	Description of the diagnosis
diagnosis.onset_date	Diagnosis/DiagOnsetDate	DG1.5	Condition.onsetDateTime	dateTime	O	Onset date of the diagnosis
diagnosis.resolution_date	Diagnosis/DiagResDate	--	Condition.abatementDateTime	dateTime	O	Resolution date of the diagnosis
diagnosis.is_primary	Diagnosis/DiagPrimary	DG1.15	EpisodeOfCare.diagnosis.rank	JSON: boolean XML: int HL7: int FHIR JSON: positiveInt* <i>*NOTE: this is mapped in the diagnosis.condition.reference section</i>	R*	Primacy Diagnosis Code indicator *Required if ICD10 is sent NOTE: if a primary indicator is not sent for any diagnoses, Dragonfly Health will default the first diagnosis in the message as the primary.
Allergies (repeatable)						
allergy.code	Allergies/Allergy/Code	AL1.2	AllergyIntolerance.code	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept* <i>*NOTE: map to the coding.code element & set coding.system to the URI to specify the code type</i>	O	Code that identifies the allergy or intolerance If no allergies are sent the entire container can be omitted. While allergies are NOT required in any ADT messages, medications cannot be dispensed without an allergy or NKA present.
allergy.description	Allergies/Allergy/Description	AL1.3	AllergyIntolerance.code	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept*	O	Allergy name



Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
				*NOTE: map to the text element		
--	Allergies/Allergy/Class	--	AllergyIntolerance.category	string	O	Allergy Category: Food Medication Environment Biologic
allergy.code_type	Allergies/Allergy/CodeType	--	--	string	O	Allergy Code Type: Ndc Gcn Gpi RxCui Text GcnSeqNo
allergy.onset_date	Allergies/Allergy/OnsetDate	AL1.6	AllergyIntolerance.onsetDateTime	dateTime	O	When allergy or intolerance was identified
allergy.resolution_date	Allergies/Allergy/ResDate	--	--	dateTime	O	The resolution date of the allergy
Primary Doctor						
doctor.last_name	Doctor/DrLastName	PV1.7.2	Practitioner.name.family	string	R	Last name of the primary doctor
doctor.first_name	Doctor/DrFirstName	PV1.7.3	Practitioner.name.given[0]	string	R	First name of the primary doctor
doctor.dea	Doctor/DrDEA	--	Practitioner.identifier	JSON: string XML: string HL7: string FHIR JSON: Identifier* *NOTE: map to the value element & set system to the URI for identifier (specifying 'DEA')	O	DEA number of the primary doctor
doctor.npi	Doctor/DrNPI	PV1.7	Practitioner.identifier	JSON: string XML: string HL7: string FHIR JSON: Identifier* *NOTE: map to the value element & set system to the URI for identifier (specifying 'NPI')	R	NPI License Number Code for primary doctor
Facility (if applicable*)						
facility.id	Facility/FacilityID	PID.11.23	Location.identifier	JSON: string XML: string HL7: string FHIR JSON: Identifier*	R*	EMR vendor internal LTC facility ID *Required for LTC/ALF pharmacy patients

Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
				*NOTE: map to the value element & set system to the URI for identifier		
facility.name	Facility/FacilityName	--	Location.name	string	O	Name of the LTC facility
facility.phone	Facility/FacilityPhone	--	Location.telecom	JSON: string XML: string HL7: string FHIR JSON: ContactPoint* *NOTE: map to the value element, set system to 'phone' & use to 'work'	O	Patient's LTC facility phone number
facility.street_address	Facility/FacilityAddress1	--	Location.address.line[0]	string	R*	Patient's LTC facility address line 1 *Required for LTC/ALF pharmacy patients
facility.address2	Facility/FacilityAddress2	--	Location.address.line[1]	string	O	Patient's LTC facility address line 2
facility.city	Facility/FacilityCity	--	Location.address.city	string	R*	Patient's LTC facility City *Required for LTC/ALF pharmacy patients
facility.state	Facility/FacilityState	--	Location.address.state	string	R*	Patient's LTC facility state *Required for LTC/ALF pharmacy patients
facility.zip	Facility/FacilityZip	--	Location.address.postalCode	string	R*	Patient's LTC facility Postal Code *Required for LTC/ALF pharmacy patients
Next of Kin / Contacts (repeatable)						
contact.last_name	Contact/ContactLastName	NK1.2.1	Patient.contact.name.family	string	O	Next of kin last name
contact.first_name	Contact/ContactFirstName	NK1.2.2	Patient.contact.name.given[0]	string	O	Next of kin first name
contact.middle_name	Contact/ContactMiddleName	NK1.2.3	Patient.contact.name.given[1]	string	O	Next of kin middle name
contact.relationship	Contact/Relationship	NK1.3.2	Patient.contact.relationship	JSON: string XML: string HL7: string FHIR JSON: CodeableConcept*	O	Next of kin relationship - Preferred relationship value examples: - Caregiver - Child - Immediate Family



Field				Datatype	Required	Description/Comments
JSON	XML	HL7	FHIR JSON			
				<i>*NOTE: map to the text element & set coding.system to the URI to specify the code type</i>		- Parent - Spouse - Self - Other - Unknown
contact.street_address	Contact/ContactAddress1	NK1.4.1	Patient.contact.address.line[0]	string	O	Next of kin address1
contact.address2	Contact/ContactAddress2	NK1.4.2	Patient.contact.address.line[1]	string	O	Next of kin address2
contact.city	Contact/ContactCity	NK1.4.3	Patient.contact.address.city	string	O	Next of kin city
contact.state	Contact/ContactState	NK1.4.4	Patient.contact.address.state	string	O	Next of kin state
contact.zip	Contact/ContactZip	NK1.4.5	Patient.contact.address.postalCode	string	O	Next of kin zip
contact.phone	Contact/ContactPhone	NK1.5.1	Patient.contact.telecom	JSON: string XML: string HL7: string FHIR JSON: ContactPoint* <i>*NOTE: map to the value element, set system to 'phone' & use to 'home'</i>	O	Next of kin phone

7. INTERFACE CONNECTION AND AUTHENTICATION

Dragonfly Health's interface connection and authentication follows industry standard processes.

7.1 Connection Protocol

HTTPS (HTTP over SSL or HTTP Secure) is Transport Layer Security (TLS) 1.2 or higher as a sub layer under regular HTTP application layering. HTTPS encrypts and decrypts user page requests as well as the pages that are returned by the Web server.

7.2 Credentials and URLs

The credentials (username and password) as well as the channel number and the ending of the URL will be provided to the vendor.

- Dragonfly Health TEST1 e.g.
 - <https://testclientservices.dragonflyhealth.com/SIG.API/chm124/PatientServices/SendAdt>
 - <https://testclientservices.dragonflyhealth.com/SIG.API/chm124/PatientServices/SendMed>
- Dragonfly Health TEST3 e.g.
 - <https://testclientservices.dragonflyhealth.com/SIG.API.3/chm124/PatientServices/SendAdt>
 - <https://testclientservices.dragonflyhealth.com/SIG.API.3/chm124/PatientServices/SendMed>
- Dragonfly Health PROD e.g.
 - <https://clientservices.dragonflyhealth.com/SIG.API/chm124/PatientServices/SendAdt>
 - <https://clientservices.dragonflyhealth.com/SIG.API/chm124/PatientServices/SendMed>

8. INTERFACE ACKNOWLEDGEMENT MESSAGES

8.1 Basic Acknowledgements

- Dragonfly Health will send an acknowledgment message when an ADT message is received and processed.
- Typical message responses:
 - Success
 - Patient not Admitted
 - Patient already discharged/deceased
 - Org not found
- Please see below the sample reply message format

9. INTERFACE SUPPORT

The EMR vendor agrees to participate in ongoing support of the interface with Dragonfly Health and being an active partner in resolving all issues to the mutual client's satisfaction.

9.1 Vendor to Vendor Meetings

The EMR vendor agrees to participate in ongoing scheduled meetings with Dragonfly Health to review upcoming interface implementations as well as review open production issues with the interface.

9.2 Client Meetings

The EMR vendor agrees to participate, in conjunction with Dragonfly Health, mutual client meetings to review interface implementation and/or interface production issues.

10. INTERFACE SERVICE LEVEL

The EMR vendor agrees to a service level of all interfaces functioning 24 hours a day/7 days a week/365 days a year.

Dragonfly Health requires an operational support escalation path with the EMR prior to going live with a client.

11. FAULT TOLERANCE

11.1 Outages

Any interface outages will be reported by Dragonfly Health at the beginning and the end of the outage. The EMR vendor agrees to have proper alerting in place to notify their internal departments and Dragonfly Health of the outage, in accordance with the Service Level Agreement. Likewise, Dragonfly Health agrees to do the same in return, in accordance with the Service Level Agreement.

In case of a time out Dragonfly Health is expecting the messages to be attempted to be sent again in a relatively short period of time. (e.g. After 10 seconds with 5 retries)

In case of a connectivity outage Dragonfly Health is expecting the messages to queue up on the client's / vendor's end. Once the connectivity is re-established all the messages in the queue should automatically be transmitted. If there is a human intervention required on the client's or vendor's end to resend failed messages, such concepts and processes need to be defined and signed off ahead of time.

11.2 Maintenance

The EMR vendor will notify Dragonfly Health and mutual clients prior to any maintenance which will require the interface to be disabled.

12. TESTING

- It is assumed that:
 - The vendor has a testing environment available for testing purposes.
 - Connectivity can be established between the vendor's testing environment and Dragonfly Health test environments specified in item #7.
 - A vendor resource will be available to perform patient actions that will trigger outbound ADT messages.

- A detailed test plan will be shared with the vendor before the testing stage of the project.

13. SAMPLE MESSAGES

13.1 JSON ADT Message

```
{
  "patient": {
    "institution": {
      "id": "51134",
      "name": "Example Institution"
    },
    "group": "ExampleGroup",
    "location": "LTC",
    "team": "",
    "pending_date": "2026-03-28 13:19:24.37",
    "admit_date": "2026-03-29 13:19:24.37",
    "location_effective_date": "2026-03-29 13:19:24.37",
    "discharge_date": null,
    "deceased": false,
    "patient_id": "1234567",
    "patient_ssn": "124-00-0031",
    "last_name": "PatientLastName",
    "first_name": "PatientFirstName",
    "middle_name": "PatientMiddleName",
    "street_address": "512 Elmwood Ave",
    "address2": "",
    "city": "Sharon Hill",
    "state": "PA",
    "zip": "19079",
    "phone": "555-867-5309",
    "birth_date": "1991-05-01",
    "sex": "F",
    "diagnosis": [
      {
        "id": "131362",
        "code": "Q22.9",
        "description": "Congenital malformation of tricuspid valve, unspecified",
        "onset_date": "2026-04-10",
        "resolution_date": null,
        "is_primary": true
      },
      {
        "id": "131012",
        "code": "R46.1",
        "description": "Bizarre personal appearance",
        "onset_date": "2026-04-10",
        "resolution_date": "2026-04-16",
        "is_primary": false
      }
    ],
    "allergy": [
      {
        "id": "123",
        "code": "00591034701",
```

```

        "code_type": "ndc",
        "description": "hydroCHLORothiazide",
        "onset_date": "2025-08-08",
        "resolution_date": null
    },
    ],
    "contact": [
        {
            "id": "12",
            "relationship": "Spouse",
            "last_name": "ExampleLastName",
            "first_name": "ExampleFirstName",
            "street_address": "512 Elmwood Ave",
            "address2": "",
            "city": "Sharon Hill",
            "state": "PA",
            "zip": "19079",
            "phone": "555-867-5309"
        }
    ],
    "doctor": {
        "last_name": "ProviderLastName",
        "first_name": "ProviderFirstName",
        "npi": "1164000371",
        "dea": "12345"
    },
    "facility": {
        "id": "1238248896",
        "name": "Example Facility Name",
        "street_address": "512 Elmwood Ave",
        "address2": "",
        "city": "Sharon Hill",
        "state": "PA",
        "zip": "19079",
        "phone": "555-867-5309"
    }
}

```

13.2 JSON ADT Message Successful Response

```

{
  "TrackingNumber": "20260430094216067957",
  "TransactionDate": "2026-04-30T09:42:16.6555718-04:00",
  "PatientId": "1234567",
  "PharmacyPatientId": "14099494",
  "Success": true,
  "Errors": []
}

```

13.3 JSON ADT Message Failure Response

```

{
  "TrackingNumber": "20260430094306678bea",
  "TransactionDate": "2026-04-30T09:43:06.8446167-04:00",
  "PatientId": "777556500",
  "PharmacyPatientId": "14099494",
  "Success": false,
  "Errors": [
    {
      "ErrorCode": "",

```

```

    "ErrorDesc": "Error|104|Patient is already discharged or deceased."
  },
  {
    "ErrorCode": "316",
    "ErrorDesc": "Patient is already discharged or deceased."
  }
]
}

```

13.4 HL7 ADT Message

```

MSH|^~\&|SendingApplication|1234|DragonflyHealth|DragonflyHealth|||ADT^A01|199||P|2.5
EVN|A01|20260324
PID|||123456^MRN||PatientLastName^PatientFirstName||19520329|F|||512 Elmwood
Ave^^Sharon Hill^PA^19079|(215) 937-5500||||364000111|||||
PV1||O|R^^^^^Home^^^Home|||1374433555^JONES^TINA|1376633248^JONES^TINA|||||Home|
|||||Identifier||||20260324||
NK1||NextOfKinLastName^NextOfKinFirstName|E^NextOfKinRelationship512 Elmwood
Ave^^Sharon Hill^PA^19079|(262) 370-1472
OBX|1|ST|Height(in)||68
OBX|2|ST|Weight(lb)||141
AL1|1|^erythromycin||20260408183116
AL1|2|^penicillins||20260408183116
DG1|1|ICD10|C53.8^Malignant neoplasm of overlapping sites of cervix
uteri^ICD10||20260424|A|||||||1

```

13.5 HL7 ADT Message Successful Response

```

MSH|^~\&|DragonflyHealth|DragonflyHealth||1234|20260430034453||ACK^A01||T|2.5
MSA|AA||Application Accept

```

13.6 HL7 ADT Message Failure Response

```

MSH|^~\&|DragonflyHealth|DragonflyHealth||1234|20260430034643||ACK^A03||T|2.5
MSA|AE||Application Error Code: 104 CodeDetail: 316 Desc: Patient is already
discharged or deceased.

```

13.7 XML ADT Message

```

<ADT>
  <Record>
    <RecIdentifier>544926:1820725</RecIdentifier>
    <RecType>10</RecType>
    <HospiceID>12345</HospiceID>
  </Record>
  <Patient>
    <PatSSN>123456789</PatSSN>
    <PatLastName>PatientLastName</PatLastName>
    <PatFirstName>PatientFirstName</PatFirstName>
    <PatMiddleName/>
    <PatAddress1>512 Elmwood Ave</PatAddress1>
    <PatAddress2/>
    <PatCity>Sharon Hill</PatCity>
    <PatState>PA</PatState>
    <PatZip>19079</PatZip>
    <PatHomePhone>(584) 202-1671</PatHomePhone>
    <PatBirthDate>04/21/1931</PatBirthDate>
    <PatGender>F</PatGender>
    <PatEthnCode/>
  </Patient>
</ADT>

```

```

        <PatHeight/>
        <PatWeight/>
        <PatMarStatus/>
        <PatID>123456</PatID>
    </Patient>
    <Hospice>
        <HospName>Example Hospice</HospName>
        <HospGroupID/>
        <HospLocType>Home</HospLocType>
        <HospTeamName>TeamA</HospTeamName>
        <HospLocEffDate>03/24/2026</HospLocEffDate>
    </Hospice>
    <Coverage>
        <CovStartDate>03/24/2026</CovStartDate>
        <CovStopDate/>
    </Coverage>
    <Diagnosis>
        <DiagICD10>Q22.9</DiagICD10>
        <ICD10Description/>
        <DiagOnsetDate/>
        <DiagResDate/>
        <DiagPrimary/>
    </Diagnosis>
    <Allergies>
        <Allergy>
            <Description>No Known Drug Allergy</Description>
            <Code>0000007055</Code>
            <OnsetDate>04/20/2026</OnsetDate>
            <ResDate/>
        </Allergy>
    </Allergies>
    <Doctor>
        <DrLastName>PhysicianLastName</DrLastName>
        <DrFirstName>PhysicianFirstName</DrFirstName>
        <DrDEA/>
        <DrNPI/>
        <DrAddress1>350 Isaacs Lane</DrAddress1>
        <DrAddress2/>
        <DrCity>Richmond</DrCity>
        <DrState>KY</DrState>
        <DrZip>40475</DrZip>
        <DrZip4/>
        <DrOfficePhone>(859) 626-9292</DrOfficePhone>
        <DrOfficeFax>(859) 626-9272</DrOfficeFax>
    </Doctor>
    <Facility>
        <FacilityId>123456</FacilityId>
        <FacilityName>Example Facility Name</FacilityName>
        <FacilityAddress1>350 Isaacs Lane</FacilityAddress1>
        <FacilityAddress2/>
        <FacilityCity>Richmond</FacilityCity>
        <FacilityState>KY</FacilityState>
        <FacilityZip>40475</FacilityZip>
        <FacilityZip4>2824</FacilityZip4>
        <FacilityPhone>(859) 626-9292</FacilityPhone>
        <FacilityFax/>
    </Facility>
</ADT>

```

13.8 XML ADT Message Successful Response

```
<Response>
  <Status>Success</Status>
  <Message>ADT message processed successfully.</Message>
  <TrackingNumber>202604301145342353da</TrackingNumber>
  <MessageControlId>087611333531803375</MessageControlId>
  <TransDate>4/30/2026 3:45:41 PM</TransDate>
</Response>
```

13.9 XML Message Failure Response

```
<Response>
  <Status>Failure</Status>
  <Message>Error|104|Patient is already discharged or deceased.</Message>
  <TrackingNumber>20260430114801788a86</TrackingNumber>
  <MessageControlId>087630210231803695</MessageControlId>
  <TransDate>4/30/2026 3:48:02 PM</TransDate>
</Response>
```

13.10 FHIR JSON ADT Message

```
{
  "resourceType": "EpisodeOfCare",
  "id": "example-episode",
  "contained": [
    {
      "resourceType": "Organization",
      "id": "example-organization",
      "identifier": [
        {
          "system":
"https://dragonflyhealth.com/identifier/organization-id",
          "value": "49414"
        }
      ],
      "name": "Example Hospice Name"
    },
    {
      "resourceType": "CareTeam",
      "id": "example-careTeam",
      "name": "Dream 1",
      "period": {
        "start": "2026-03-31T06:49:00.277Z"
      }
    },
    {
      "resourceType": "Patient",
      "id": "example-patient",
      "identifier": [
        {
          "system": "https://exampleVendor.com/identifier/mrn",
          "value": "629360331"
        },
        {
          "system": "http://hl7.org/fhir/sid/us-ssn",
          "value": "123456789"
        }
      ]
    }
  ],
}
```

```

"name": [
  {
    "use": "official",
    "family": "ExampleLastName",
    "given": [
      "ExampleFirstName",
      "ExampleMiddleName"
    ]
  }
],
"gender": "male",
"birthDate": "1963-08-18",
"address": [
  {
    "use": "home",
    "type": "both",
    "line": [
      "2233 TEST Tree Forest Circle"
    ],
    "city": "Dallas",
    "state": "TX",
    "postalCode": "75248",
    "country": "USA"
  }
],
"telecom": [
  {
    "system": "phone",
    "value": "(372)-410-3584",
    "use": "home",
    "rank": 1
  }
],
"contact": [
  {
    "relationship": [
      {
        "coding": [
          {
            "system":
"http://terminology.hl7.org/CodeSystem/v2-0131",
            "code": "S"
          }
        ],
        "text": "Spouse"
      }
    ],
    "name": {
      "family": "ExampleContactLastName",
      "given": [
        "ExampleContactFirstName"
      ]
    },
    "telecom": [
      {
        "system": "phone",
        "value": "555-555-5555"
      }
    ]
  }
],

```

```

        "address": {
            "line": [
                "2233 TEST Tree Forest Circle"
            ],
            "city": "Dallas",
            "state": "TX",
            "postalCode": "75248"
        }
    ],
},
{
    "resourceType": "Practitioner",
    "id": "example-practitioner",
    "active": true,
    "identifier": [
        {
            "system": "http://hl7.org/fhir/sid/us-npi",
            "value": "3235655611"
        },
        {
            "system": "http://hl7.org/fhir/sid/us-dea",
            "value": "123456"
        }
    ],
    "name": [
        {
            "use": "official",
            "family": "SampleMDLastName",
            "given": [
                "SampleMDFirstName"
            ]
        }
    ]
},
{
    "resourceType": "AllergyIntolerance",
    "id": "example-allergy",
    "type": "allergy",
    "category": [
        "environment"
    ],
    "code": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "300916003",
                "display": "Latex"
            }
        ],
        "text": "Latex"
    },
    "onsetDateTime": "2021-03-24T06:49:00.277Z",
    "subject": {
        "reference": "#example-patient"
    }
},
{
    "resourceType": "Condition",

```

```

      "id": "example-condition",
      "code": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/sid/icd-10-cm",
            "code": "G12.21"
          }
        ],
        "text": "Amyotrophic Lateral Sclerosis"
      },
      "onsetDateTime": "2021-02-04T00:00:00-06:00",
      "subject": {
        "reference": "#example-patient"
      }
    },
    {
      "resourceType": "Location",
      "id": "example-facility",
      "identifier": [
        {
          "system":
"https://exampleVendor.com/identifier/facilityId",
          "value": "FAC111048"
        }
      ],
      "name": "Example Facility",
      "telecom": [
        {
          "use": "work",
          "value": "2705295555",
          "system": "phone"
        }
      ],
      "address": {
        "use": "office",
        "line": [
          "460 S College St"
        ],
        "state": "KY",
        "postalCode": "42170-9738",
        "city": "Woodburn"
      }
    }
  ],
  "status": "active",
  "diagnosis": [
    {
      "condition": {
        "reference": "#example-condition"
      },
      "rank": 1
    }
  ],
  "patient": {
    "reference": "#example-patient"
  },
  "managingOrganization": {
    "reference": "#example-organization"
  },

```

```
    "period": {  
      "start": "2021-03-31T06:49:00.277Z"  
    }  
  }  
}
```

13.11 FHIR JSON ADT Message Successful Response

```
{  
  "resourceType": "OperationOutcome",  
  "issues": {  
    "severity": "success",  
    "details": "Id|97265402; Success|Id represents PharmacyPatientId"  
  }  
}
```

13.12 FHIR JSON ADT Message Failure Response

```
{  
  "resourceType": "OperationOutcome",  
  "issues": {  
    "severity": "error",  
    "details": " Error|104|Patient is already discharged or deceased."  
  }  
}
```